

8.5 OTHER TECHNOLOGIES

Thermo Fisher's ion semiconductor sequencing technology-based platforms include Ion Torrent PGM, Ion Proton, and the newly launched GeneStudio S5 Series. Ion semiconductor sequencing involves the direct translation of genetic codes of sequence (DNA bases) into digital information onto a semiconductor chip. The technology translates chemically encoded information (A, C, G, T) into digital information (0, 1) on a semiconductor chip. The company focuses on continuous enhancements in the technology to widen its scope of applications and increase its precision of sequencing.

The launch of new and advanced sequencers by Thermo Fisher and the advantages offered by ion semiconductor technology platforms are expected to drive market growth in the near future. For instance, in March 2022, the company launched the Ion Torrent Genexus Dx System. This fully validated system enables users to perform both diagnostic testing and clinical research on a single instrument. The company also provides complete NGS workflow for diagnostic solutions for comprehensive genomic profiling. The high adoption of ion semiconductor technology can be attributed to its features, including simpler workflow, time- as well as cost-efficiency, and scalability. This technology is gaining popularity in microbial profiling, thus contributing to the overall segment growth.

TABLE 66 MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	46.18	52.10	58.75	66.31	74.91	84.91	96.58	110.34	13.4%
Europe	25.14	28.51	32.32	36.67	41.63	47.44	54.24	62.28	14.0%
Asia Pacific	6.96	8.08	9.36	10.85	12.58	14.62	17.04	19.94	16.3%
Latin America	1.59	1.75	1.91	2.09	2.29	2.52	2.77	3.07	9.9%
Middle East and Africa	1.10	1.19	1.29	1.39	1.50	1.63	1.76	1.91	8.1%
Total	80.97	91.63	103.64	117.32	132.92	151.11	172.40	197.53	13.8%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 67 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	41.85	47.25	53.33	60.24	68.11	77.27	87.96	100.57	13.5%
Canada	4.33	4.85	5.42	6.07	6.80	7.65	8.62	9.77	12.5%
Total	46.18	52.10	58.75	66.31	74.91	84.91	96.58	110.34	13.4%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 68 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	5.43	6.18	7.04	8.02	9.15	10.47	12.02	13.86	14.5%
UK	4.86	5.56	6.35	7.27	8.32	9.56	11.03	12.76	15.0%
France	3.84	4.36	4.95	5.62	6.39	7.28	8.34	9.58	14.1%
Italy	1.08	1.23	1.39	1.57	1.78	2.02	2.30	2.64	13.7%
Spain	0.83	0.94	1.06	1.20	1.35	1.53	1.74	1.99	13.4%
Rest of Europe	9.09	10.24	11.53	12.99	14.65	16.57	18.81	21.44	13.2%
Total	25.14	28.51	32.32	36.67	41.63	47.44	54.24	62.28	14.0%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 69 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	2.94	3.42	3.96	4.60	5.34	6.21	7.25	8.49	16.4%
Japan	1.37	1.60	1.86	2.17	2.53	2.96	3.47	4.08	17.0%
India	0.17	0.20	0.23	0.27	0.31	0.36	0.42	0.49	16.0%
Rest of Asia Pacific	2.48	2.86	3.30	3.81	4.40	5.09	5.90	6.88	15.8%
Total	6.96	8.08	9.36	10.85	12.58	14.62	17.04	19.94	16.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 70 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR OTHER TECHNOLOGIES, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.75	0.82	0.90	0.99	1.08	1.19	1.31	1.45	10.0%
Rest of Latin America	0.84	0.93	1.01	1.11	1.21	1.33	1.46	1.61	9.8%
Total	1.59	1.75	1.91	2.09	2.29	2.52	2.77	3.07	9.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis